C Programming Assignment – Question 3

3. Treasure Hunt

Question

Blackbeard and three other pirates find gold coins. Long Ben receives x% of the treasure. Blackbeard then receives y% of the remaining treasure. The other three pirates share the remaining whole coins equally. Any leftover coins are thrown into the sea.

Sample Input

```
729
65
87
```

Sample Output

```
473
222
11
```

C Program

```c
#include <stdio.h>

int main(void) {
    int treasure, x, y;
    int longBen, remaining, blackbeard, pirateShare;

    printf("Enter total treasure: ");
    scanf("%d", &treasure);
    printf("Enter Long Ben's percentage: ");
    scanf("%d", &x);
    printf("Enter Blackbeard's percentage: ");
    scanf("%d", &y);

    longBen = treasure * x / 100;
    remaining = treasure - longBen;

    blackbeard = remaining * y / 100;
    remaining -= blackbeard;

    pirateShare = remaining / 3;

    printf("%d\n%d\n%d\n", longBen, blackbeard, pirateShare);
    return 0;
}
```

Explanation

1. Calculate Long Ben's share as x% of the total treasure.

2. Subtract Long Ben's share from the treasure.

3. Calculate Blackbeard's share as y% of the remaining treasure.

4. Divide the remaining coins equally among the three other pirates using integer division.

5. Print Long Ben's share, Blackbeard's share, and each other pirate's share.

Note

Integer division drops fractional coins, matching the rule that leftover coins are discarded.